



BCSE498J- Project - II

DUAL-DOMAIN LEARNING FRAMEWORK FOR ECG (DD-ECGNET)

Submitted in partial fulfilment of the requirements for the degree of

Bachelor of Technology

in

Computer Science and Engineering

by

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April, 2026

DECLARATION

I hereby declare that the project entitled “DUAL-DOMAIN LEARNING FRAMEWORK FOR ECG (DD-ECGNET)” submitted by me, for the award of the degree of Bachelor of Technology in Computer Science and Engineering to VIT is a record of bonafide work carried out by me under the supervision of Prof. Manjula R, School of Computer Science and Engineering.

I further declare that the work reported in this project report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Vellore **DEVABHAKTUNI LOHITHA PADMAVATHI (22BCE0309)**

Date :

CERTIFICATE

This is to certify that the project entitled “DUAL-DOMAIN LEARNING FRAMEWORK FOR ECG (DD-ECGNET)” submitted by DEVABHAKTUNI LOHITHA PADMAVATHI (22BCE0309), School of Computer Science and Engineering, VIT, for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is a record of bonafide work carried out by the student under my supervision during Winter Semester 2025-2026, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The project fulfills the requirements and regulations of the University and in my opinion meets the necessary standards for submission.

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EXECUTIVE SUMMARY

This project proposes a combined deep learning system for automatic heartbeat classification from electrocardiograms (ECGs) using MIT-BIH arrhythmia database. The aim of this classifier system is to detect various types of cardiac arrhythmias using electrocardiograms (ECGs) taking into account issues like imbalanced classes, inter-patient variability, and physiological noise in the recordings.

In our work, a combination of morphological and rhythmical aspects is employed. Small and asymmetric beat windows with regard to R-peak in ECG recordings are used for morphology, whereas HRV features based on the analysis of RR intervals represent rhythm information. Preprocessing of ECGs involves filtering and noise removal prior to feature extraction.

For better classification results, a combined model was constructed in which CNN is used to extract spatial characteristics of the ECG waveforms, and recurrent layers serve for sequential learning of rhythm-based temporal features. For this purpose, RR interval data are fed into a separate dense network branch and merged with ECG waveform representations at the end. Training of this hybrid model relies on Focal loss due to imbalanced nature of the dataset

The dataset was analyzed through Stratified Group 10-Fold Cross Validation, which guarantees that there is no repetition of patient entries from training to testing datasets, hence increasing evaluation accuracy and clinical validity. Other techniques like weighted sampling, gradient clipping, learning rate schedule, and model checkpoints have been employed to enhance the stability of training processes.

The system categorizes heartbeats into four types of arrhythmias, namely Normal (N), Supraventricular Ectopic Beats (SVEB), Ventricular Ectopic Beats (VEB), and Fusion Beats (F). Experimental findings reveal the significance of signal normalization, sampling balance, and loss weighting techniques for convergence and minority classes.

This study proposes a feasible framework that integrates hybrid deep learning methods for intelligent identification of cardiac arrhythmias, setting up future frameworks for clinical support systems and wearable devices for continuous ECG monitoring.

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LIST OF ABBREVIATIONS

1D-CNN	One-Dimensional Convolutional Neural Network
AAMI	Association for the Advancement of Medical Instrumentation
CNN	Convolutional Neural Network
ECG	Electrocardiogram
F	Fusion Beat
FFT	Fast Fourier Transform
HRV	Heart Rate Variability
LSTM	Long Short-Term Memory
MIT-BIH	Massachusetts Institute of Technology – Beth Israel Hospital Database
pNN50	Percentage of NN intervals differing by greater than 50 ms
RMSSD	Root Mean Square of Successive Differences
SD1	Poincaré plot short-axis measure
SD2	Poincaré plot long-axis measure
SDNN	Standard Deviation of NN intervals
SVEB	Supraventricular Ectopic Beat
SVM	Support vector machine
VEB	Ventricular Ectopic Beat

SYMBOLS AND NOTATIONS

α Weight parameter

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Cardiovascular diseases continue to remain the leading cause of death across the world, accounting for approximately 17.9 million deaths annually, according to the World Health Organisation. And around 28 per cent of all deaths that were recorded in 2023 in India are caused by CVDs. Among cardiovascular conditions, cardiac arrhythmias represent a particularly dangerous class of disorders characterised by abnormal electrical activity of the heart. Arrhythmias range from harmless ectopic beats to life-threatening conditions such as ventricular fibrillation, and their early detection is critical for timely clinical intervention. The electrocardiogram (ECG) is a non-invasive diagnostic tool for arrhythmia detection. It records the electrical activity of the heart as a voltage signal by placing electrodes on the surface of the human body. An Electrocardiogram (ECG) waveform mainly comprises the P-wave, which represents atrial depolarisation, the QRS complex, which represents ventricular depolarisation, and the T-wave, which represents ventricular repolarisation. The deviations in the amplitude, duration, or timing of these components are associated with different arrhythmia types clinically. Previously, arrhythmia diagnosis has relied only on cardiologists' interpretation of ECG recordings. However, the ambulatory ECG monitoring systems produce recordings lasting from 24 to 48 hours or more, making manual analysis inefficient and prone to human errors. This has driven sustained research interest in automated ECG classification systems that are capable of providing consistent, real-time analysis of long-duration cardiac recordings.

1.1.1 Machine Learning-based Approaches

Biomedical signals were recognised early on as one of the most potential application domains. (Jagadeep Rahul, et al). ECG is one of the important biomedical signals that can be used to detect anomalies in the shape of the heart rhythm. The early automated ECG classification systems relied heavily on rule-based systems. Later on, feature extraction was combined with conventional machine learning models for the

classification-related tasks. These approaches typically involved signal preprocessing for smoothening the PQRS waveforms and for noise removal, feature extraction from ECG segments, and classification using a trained model. Support Vector Machines (SVM) have been widely applied to arrhythmia classification (Amine ben slama, et al), with researchers extracting features such as QRS duration, PR interval, and RR interval statistics as input vectors. Principal Component Analysis (PCA) was employed in several other research papers to reduce the dimensionality of the beat features before classification (V. Gupta et al.). Wavelet-based feature extraction, like using CWT (continuous wave transform) and DWT (discrete wave transform), has become widely used due to their compatibility with ECG signals, which are non-stationary by nature. k-Nearest Neighbours (kNN), Decision Tree, Random Forest, and Linear Discriminant Analysis (LDA) classifiers have all been evaluated on the Massachusetts Institute of Technology – Beth Israel Hospital Database (MIT-BIH) Arrhythmia Database with varying levels of improvement in the evaluation metrics. But the key limitation of all the machine learning approaches is the requirement for manual feature engineering, which demands domain expert knowledge, is time-consuming, and may systematically miss information that is not properly captured by predefined features.

1.1.2 Deep Learning-based Approaches

Deep learning has successfully transformed automated ECG analysis by enabling actual learning from raw signals without manual feature engineering. Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM) networks, and their hybrid combinations have all been applied to arrhythmia classification with good results. Li et al. (2017) designed a six-layer One-Dimensional Convolutional Neural Network (1D-CNN) for arrhythmia classification on the MIT-BIH database. Acharya et al. (2017) proposed a deeper Convolutional Neural Network (CNN) with eleven layers. The adoption of residual connections and skip connections from ResNet improved ECG classification, as training of much deeper networks without the gradient descent problem has become possible. Recurrent architectures leverage the sequential and time-dependent nature of ECG signals. Gao et al. (2019) implemented an LSTM-based system addressing class imbalance, while Yildirim (2018) applied bidirectional Long Short-Term Memory (LSTM) for five-class beat classification. Petmezas

et al. (2021) proposed a hybrid CNN-LSTM model trained with Focal Loss for atrial fibrillation detection on the MIT-BIH AF database, achieving 97.87 percent sensitivity and 99.29 percent specificity. Wang et al. (2023) proposed time-frequency domain fusion using FFT-augmented CNN inputs on the MIT-BIH Arrhythmia database, reporting 99.43 percent overall accuracy. These two works serve as the primary methodological references for this project.

subsection content to be included

1.2 MOTIVATION

1.2 MOTIVATION Even though automated ECGECG classification has come a long way from being a theoretical concept to actually being used in medical labelling and related research work, there are still some important gaps in the current research that this project aims to fill. First, most of the methods that are already out there use either time-domain morphological features or frequency-domain rhythm features, but they don't often combine the two in a clear, unified way. Wang et al. (2023) combine ECGECG signals with Fast Fourier Transform (FFT) FFT frequency components at the input level; however, they neglect to include Heart Rate Variability (HRV) features, which are a significant source of global rhythm information extensively utilised in clinical cardiology. The combination of Heart Rate Variability (HRV) HRV features with deep morphological features from a CNN-BiLSTM backbone has not been investigated in relation to the MIT-BIH inter-patient arrhythmia classification benchmark. Second, there is a huge difference in the number of classes in real-world ECG databases. The Association for the Advancement of Medical Instrumentation (AAMI) classification of the MIT-BIH Arrhythmia Database shows that about 89.4 percent of beats are normal. A lot of published methods that claim to have high overall accuracy are usually because the Normal class is significantly present in the dataset, but don't actually use the required metrics to test how well they can classify the minority class. Wang et al. (2023) have given only the overall accuracy and not the F1-scores for each class, which makes it difficult to perceive how well the model works on clinically important minority arrhythmia types. Third, many published works don't use inter-patient protocols when they evaluate things. A random train-test split lets beats from the same patient show up in both the

train and test sets. This makes performance look better because the model learns from that patient instead of really generalising. Patient-level cross-validation is the correct method for determining clinical relevance.

1.3 SCOPE OF THE PROJECT

1.3 SCOPE OF THE PROJECT The scope of this project will involve developing and evaluating a hybrid Deep Learning arrhythmia classification architecture using the MIT-BIH Arrhythmia database. Within scope: The end-to-end ECG processing workflow, including loading the data from a file, preprocessing, feature extraction, model building, training and evaluation. Preprocessing inspired by Petmezas et al.: Highpass 7th-order Butterworth filter with a cutoff frequency of 0.5 Hz; Lowpass 6th-order Butterworth filter with a cutoff frequency of 40 Hz; Denoising using 3 levels of Discrete Wavelet Transform using the db4 wavelet with soft thresholding. Beat-centred windowing: 270 sample convolutional neural network (CNN) window (250 milliseconds pre R-peak + 500 ms post R-peak) and 60 seconds context window for HRV analysis per annotated beat. Nine HRV features: Standard Deviation of NN intervals (SDNN), Root Mean Square of Successive Differences (RMSSD), Percentage of NN intervals differing by greater than 50 ms (pNN50), Mean μ , σ , LF/HF, Poincaré plot short-axis measure (SD1), Poincaré plot long-axis measure (SD2). Architecture: Hybrid ResNet + BiLSTM (Morphology) architecture and Dense layer (HRV) concatenated output architecture. Imbalance handling strategies: Focal loss ($\gamma=2$), Weighted Random Sampling, Class weighting. Group Patient Stratified 10-fold cross-validation with metrics calculated per-class. Outside scope: Multichannel ECG analysis, analysis limited to the MLII lead. ECG real-time data acquisition system. Use of additional ECG databases for testing (PTB-XL, CPSC 2018, Chapman-Shaoxing)

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

2.1 LITERATURE REVIEW

2.1.1 Machine Learning Based

In recent years, machine learning (ML) algorithms have gained paramount importance for predicting and early diagnosis of heart diseases, which account for a significant number of deaths around the globe (RAJU DEVI, 2024)(Sheng, 2024)(Budiman et al., 2024)(Abdellatif et al., 2024)(Islam et al., 2024)(Parashar et al., 2024)(Al-Alshaikh et al., 2024)(Shinde et al., 2024)(Bhamare et al., 2024)(A. et al., 2023)(Yadav, 2024)(Reddy et al., 2024)(Lakshmi Satyanarayana, 2024)(Sharma et al., 2024)(Anand et al., 2024)(Ambala et al., 2024)(Pawar et al., 2024)(Wan et al., 2025)(Gao, 2024). Considering that clinical data is highly complex and computationally intensive, automated methods are needed for timely medical intervention. This review analyses the findings from recent peer-reviewed literature. It analyses the application, performance and challenges by adopting various ML algorithms in the ECG classification. The area of automatic classification of ECG arrhythmias through machine learning is not a new one. The creation of the MIT-BIH Arrhythmia Database by Moody and Mark in 2001 set a benchmark for comparing different algorithms. One of the most common techniques employed was based on the use of wavelet transforms. The discrete wavelet transform breaks down an ECG signal into different frequency bands, enabling its compact representation while also reflecting its local features and rhythms. A plethora of studies reveal the efficiency of ensemble models like Random Forest and XGBoost. For example, XGBoost was noted as a good performer with accuracies ranging from 63 percent to 85 percent in heart failure prediction (Yadav et al., 2024). Moreover, the accuracy of Random Forest classification was also noted when predicting cardiovascular diseases (Lakshmi Satyanarayana, 2024)(Anand et al., 2024). The comparison usually shows higher accuracy of models like XGBoost, LightGBM, and Random Forest compared to conventional statistical analysis and even other machine learning methods because of the capacity of such models to handle complex interaction between different variables (Anjum et al., 2024)(Yadav et al., 2024)(Elbehiery et al., 2025)(Suhatrill et al.,

2024). For example, the comparison of Logistic Regression, Support Vector Machine, XGBoost, LightGBM, and Decision Tree revealed the effectiveness of XGBoost for myocardial infarction prediction (Anjum et al., 2024). In addition, another comparative study considered the performance of eight algorithms, such as Decision Tree, Naïve Bayes, K-Nearest Neighbors, Support Vector Machine, Random Forest, Logistic Regression, Neural Network, and Gradient Boosting, in relation to a specific dataset, e.g., Framingham Heart Study data (Suhatri et al., 2024). Methods based on RR-intervals use the relative timing of successive heartbeats to classify rhythms. Since ventricular ectopic beats are typically marked by an early heartbeat followed by a compensatory pause, various measures related to RR-intervals can be very useful in the classification task. Andersen et al. (2017) used Support vector machine (SVM) with inter-beat intervals as features to detect atrial fibrillation. In a study comparing Random Forest and () methods, Kennedy et al. (2016) used various RR-intervals. Several parties have used PCA to extract beat templates. As a result, the feature vector representing the majority of morphological differences was able to be derived from the projections of individual beats in a low-dimensional subspace generated through PCA on the training samples. The task of separating ECG waveforms that were mixed was performed using ICA. The method presented by Yang et al. (2018), which utilised () for extracting features, followed by a linear SVM for classification of arrhythmia, proved to be accurate and computationally efficient. Regardless of their success, machine learning methods suffer from one fundamental flaw: necessity of engineered features.

2.1.2 Deep Learning-based

Deep learning-based approaches have mostly outperformed traditional machine learning techniques in the case of ECG classification, mainly due to automatic feature extraction hierarchies using deep neural networks. For example, Li et al. (2017) used a six-layered CNN to classify ECGs in the MIT-BIH database and proved the ability of deep convolutional neural networks to automatically learn the morphology features of ECG data. Similarly, Acharya et al. (2017) presented an advanced deep network structure consisting of three convolutional layers, three max pooling layers, and three fully-connected layers. Residual connections (ResNet, He et al., 2016) facilitated training much deeper neural networks without vanishing gradient problems. According to

Wang et al. (2023), their shallow 1D-CNN model combines ECG signal features from the time domain with FFT features from the frequency domain in the input layer. In their pre-processing, db5 wavelet transform is applied at 9 levels. For each 300 sample heart beat, they added 180 FFT samples to produce an input feature size of 480. Four layers of the CNN model with increasingly larger kernel sizes obtained 99.43 percent accuracy rate using a random 70/30 train/test split for 46 MIT-BIH records. Recurrent networks leverage the sequential nature of the ECG signal. Gao et al. (2019) employed a system based on LSTM recurrent networks for detecting heart arrhythmias on the imbalanced MIT-BIH dataset. Yildirim (2018) suggested bi-directional LSTM for classifying ECG beats into five classes, showing that bi-directional LSTM improves the classification of morphologically challenging beat types. Hybrid architectures such as CNN-LSTM models leverage spatial analysis along with temporal modeling. Petmezas et al. (2021) designed a CNN-LSTM hybrid architecture trained with Focal Loss to classify four different types of ECG rhythms from the MIT-BIH AF database. Their approach produced a sensitivity of 97.87 percent and a specificity of 99.29 percent, employing stratified 10-fold cross-validation. Through the ablation study across various gamma values (0, 0.5, 1, 2, 3), they showed that Focal Loss with gamma value of 2 provides the best sensitivity and specificity. In this project, we adopt their preprocessing pipeline, beat segmentation approach, Focal Loss, and 10-fold cross-validation design.

2.2 GAPS IDENTIFIED

- Gap 1 - Absence of HRV-CNN fusion: There has been no attempt to fuse HRV features, manually crafted and covering aspects of time domain, frequency domain, and nonlinear domains, with morphological features extracted using a CNN approach in MIT-BIH Arrhythmia Database literature so far.
- Gap 2 – Inadequate imbalance management: Wang et al. (2023) do not perform any imbalance management at all. Most of the other studies only mention an aggregate accuracy measure that depends on the majority class 'Normal' with the highest representation (89.4 percent) in the dataset.
- Gap 3 – Data leakage in testing: Wang et al. (2023) conduct a split of 70/30 randomly that permits data leaks between training and testing datasets. The re-

sulting inflated accuracy measures cannot be taken into account since there will be patient overlap during testing.

- Gap 4 – HRV features with short window: Earlier studies that used beat windows of 10 seconds were unable to derive frequency domain HRV features due to insufficient duration, because the low frequency band (0.04 to 0.15 Hz) takes a minimum of 25 seconds of RR intervals. The present study uses a 60-second context window that solves this problem.
- Gap 5 – Non AAMI classifications: According to Wang et al. (2023), there are five classifications (N, A, V, L, R) which do not conform to AAMI classification system. According to AAMI, L (Left Bundle Branch Block) and R (Right Bundle Branch Block) are classified as normal. Hence, their classification is less imbalanced compared to the standard inter-patient problem.

2.3 OBJECTIVES

- To construct an end-to-end pipeline for ECG data processing from loading to preprocessing, window selection, feature engineering, and model training based on the MIT BIH Arrhythmia Database.
- To design a novel beat-centric hybrid windowing system comprising 270-sample CNN windows and 60-second HRV context windows.
- To derive 9 HRV features per beat including time-domain, frequency-domain, and nonlinear Poincaré domain features.
- To construct a novel hybrid deep learning framework integrating 1D-ResNet and BiLSTM networks for morphology feature extraction with a HRV dense network branch.
- To tackle the issue of class imbalance using Focal Loss ($\alpha = 2$), Weighted Random Sampling, and balanced class weights for each fold.
- To conduct experiments utilizing patient-based stratified group 10-fold cross validation and report per-class precision, recall, F1-score, and macro F1-score as key performance metrics.

- To make a comprehensive comparison between our work and Wang et al. (2023) and Petmezas et al. (2021).

2.4 PROBLEM STATEMENT

The MIT-BIH arrhythmia database contains 46 records of two-channel ECGs recorded at 360Hz with annotations on the beat level made by specialists. The task here is to develop an automated deep learning model for classification of beats from annotated ECGs into four types according to AAMI standards (Normal - N, Supraventricular Ectopic Beat – SVEB, Ventricular Ectopic Beat – VEB, and Fusion – F) with consideration for extreme class imbalance between Normal (89.4 percent), VEB (7.0 percent), SVEB (2.7 percent), and Fusion (0.8 percent).

2.5 PROJECT PLAN

- Phase 1: Data acquisition and pre-processing (Weeks 1-2): Acquisition of all 46 records from MIT-BIH dataset via the WFDB Python interface. Application of the three-step Petmezas pre-processing pipeline. Signal quality verification by visual inspection of filtered signals on different records and different types of arrhythmias.
- Phase 2: Features extraction (Weeks 3-4): Windowing procedure based on annotation files (R-peak positions). Creation of the 9-feature HRV extraction pipeline with frequency-domain approach employing the Welch PSD method. HRV features values verification against physiological ranges. Final dataset construction with 97,585 labelled heartbeats with corresponding windows in ECG and HRV features.
- Model creation (Weeks 5-7): Implementation of the hybrid ResNet-BiLSTM structure. Implementation of Focal Loss function, Weighted Random Sampling, CosineAnnealingLR scheduler and gradient clipping. Hyperparameters tuning based on Petmezas et al. (2021).
- Phase 4: Training and evaluation (Weeks 8-10): Patient-based Stratified Group Cross-Validation for 10 folds. Per fold training for 100 epochs with checkpoints

every 5 epochs. The best model selection based on F1.

- Analysis and reporting (Weeks 11-12): Result processing. Comparing results with reference papers. Preparing this report.

CHAPTER 3

TECHNICAL SPECIFICATION

3.1 REQUIREMENTS

3.1.1 Functional Requirements

- FR1 - Signal Loading: The system will load ECG signals from the MIT-BIH Arrhythmia Database using WFDB and extract the MLII signal (channel number = 0) as a float32 array sampled at 360 Hz.
- FR2 - Annotation Loading: The system will load annotations in beat level using the WFDB atr file format and extract the sample indices of R-peaks and beat types for all 46 databases.
- FR3 - Pre-processing: Three-step process – (a) seventh order Butterworth high-pass filter at 0.5 Hz; (b) sixth order Butterworth low-pass filter at 40 Hz; (c) three-level discrete wavelet transform using db4 with soft-threshold denoising.
- FR4 - AAMI Classification: All annotated beats will be classified into four classes according to AAMI guidelines (N=0, SVEB=1, VEB=2, F=3) ignoring paced and unknown beats.
- FR5 - Windowing: Per-beat CNN input windows of 270 samples (90 pre R-peak and 180 post R-peak) and HRV context windows of 21,600 samples (60 seconds) centred around each annotated beat.
- FR6 - HRV Calculation: Nine HRV features calculated for each beat from RR interval series within the 60-second context window: SDNN, RMSSD, pNN50, mean RR, LF power, HF power, LF/HF ratio, SD1, and SD2.
- FR7 - ECG Normalisation: Each ECG window of size 270 samples normalised to zero mean and unit variance.
- FR8 - HRV Normalisation: Features computed from HRV normalised using StandardScaler trained only on the training fold.

- FR9 - Data Balance: Weighted Random sampling of samples with weight assigned based on the inverse frequency of the class in the training fold.
- FR10 - Model Training: Hybrid CNN-BiLSTM-HRV model trained with Focal loss ($\gamma = 2$), class-balanced weights, Adam optimiser and CosineAnnealingLR learning schedule.
- FR11 - CV Procedure: Stratified Group K-Fold cross-validation with 10-folds and groups determined based on patient ID; no patient belongs to both train and test set.
- FR12 - Performance Report: Report per-class precision, recall, f1-score and macro f1-score per fold along with its mean and standard deviation.

3.1.2 Non-Functional Requirements

- NFR1 - Reproducibility: All random seeds set ($random_state = 42$) during cross-validation, split creation, model initialization, and data sampling processes.
- NFR2 - Scalability: Entire pipeline from preprocessing to training successfully executed within realistic timelines on cloud-based GPU-enabled infrastructure.
- NFR3 - Modularity: Architecture organized into distinct phases that can be separately altered and validated.
- NFR4 - Compliance with standards: Classification of beat types according to class definitions established by AAMI EC57 standard.
- NFR5 - Absence of data leakage in evaluation: Patient's data shall not be present in both train and test datasets in any fold.

3.2 FEASIBILITY STUDY

3.2.1 Technical Feasibility

The project is viable from a technical perspective. First, the MIT-BIH Arrhythmia Dataset is freely available via PhysioNet and can be accessed using the WFDB Python package. Second, all necessary packages to run the code (PyTorch, WFDB, PyWavelets, SciPy, scikit learn) are open-source and compatible with Python version 3.12. Third,

the full size of the dataset (97,585 beats, 270 samples per beat with float32 data type) is 100 MB, which will easily fit into memory on both GPU and CPU machines. Fourth, GPU-based training can be done with CUDA through various cloud solutions.

3.2.2 Economic Feasibility

The cost involved in this project is low. The MIT-BIH Arrhythmia Database can be accessed freely through PhysioNet’s open-access database license. The software libraries used are all open-sourced and free of any licensing costs. Compute resources are offered through GPU cloud computing facilities that are either free or cheap. The major expense lies in the research effort.

3.2.3 Social Feasibility

The project tackles an important problem that has significant clinical importance with public health repercussions. The system for automatically detecting arrhythmias can relieve cardiologists of the burden of evaluating long-term Holter monitor recordings by presenting only sections that have been highlighted as problematic. The use of class definitions based on AAMI standards ensures that the results obtained are clinically relevant.

3.3 SYSTEM SPECIFICATION

3.3.1 Hardware Specification

Table 3.1: Hardware Specification

Component	Specification
GPU	NVIDIA GPU (CUDA-enabled) using cloud environment
CPU	Multi-core Intel or AMD processor
RAM	16 GB minimum (32 GB recommended)
Storage	10 GB free disk space
Network	Internet access for MIT-BIH download using PhysioNet API

3.3.2 Software Specification

Table 3.2: Software Specification

Library	Version	Purpose
Python	3.12	Primary programming language
PyTorch	2.x	Deep learning models and GPU training
WFDB	4.3.1	MIT-BIH data and annotation access
PyWavelets	1.x	DWT preprocessing (db4 wavelet)
SciPy	1.17.x	Filtering, Welch PSD, interpolation
NumPy	1.26.x	Array operations and feature computation
scikit-learn	1.x	Cross-validation, scaling, metrics

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

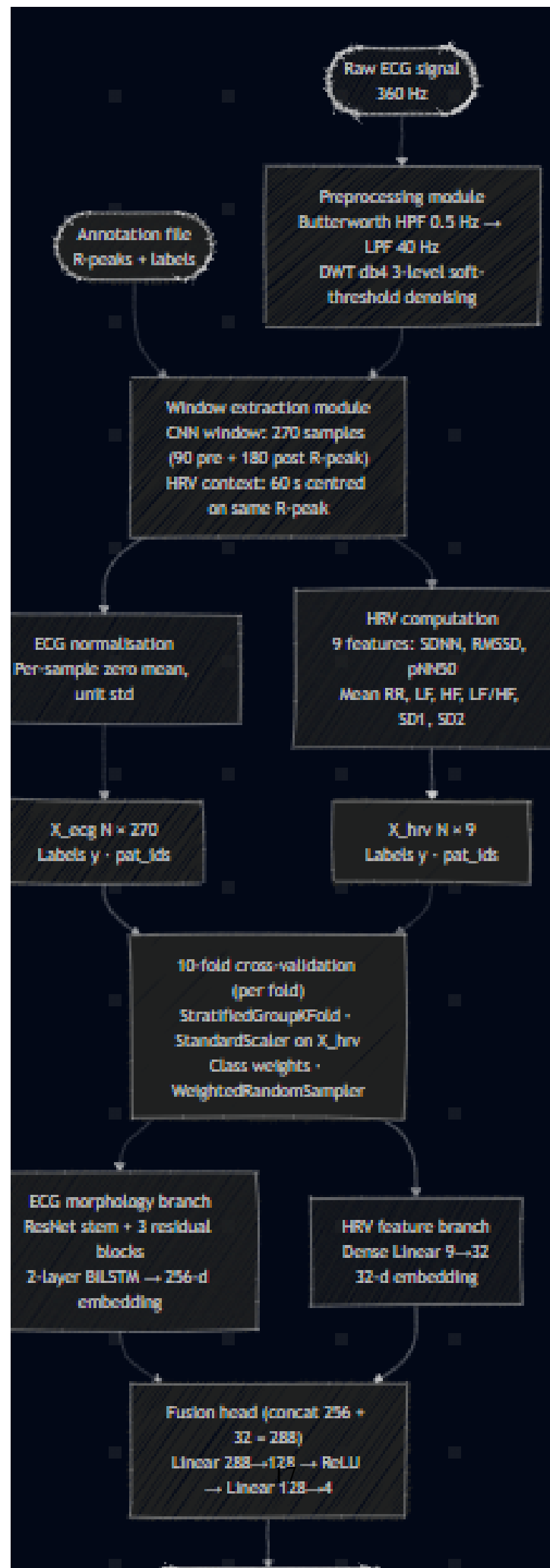
The architecture itself consists of a pipeline of five stages: data ingestion, preprocessing, feature extraction, model inference, and evaluation. The pipeline itself is designed to be modular, with each stage being completely independent from the others in terms of its input and output. This means that signal-level processing (which is performed with the help of NumPy, SciPy, and PyWavelets) can be decoupled from learning-level processing (performed with the help of PyTorch). Firstly, data ingestion loads raw ECG signals and their annotations from the MIT-BIH database using the WFDB interface. Secondly, preprocessing performs the three-stage Petmezas filtering algorithm on all 46 records and provides a collection of clean ECG signals stored as float32 arrays in memory. Thirdly, feature extraction uses a hybrid windowing technique which creates 270-point ECG snippets together with their 9 dimensional HRV feature vectors corresponding to the annotated beats. Fourthly, model implementation involves the use of dual branch architecture. Fifthly, evaluation is based on the 10-fold patient-level cross-validation together with metrics reporting and cross-validation folds aggregation. Dual-branch fusion architecture is the most distinctive architectural innovation. The ECG branch uses deep learning to represent local beat morphology, whereas the HRV branch uses manually crafted features for representing global rhythm patterns. The combination of this complementary information with a dense classifier will be useful for the model to leverage spatial, temporal and rhythmic discriminative information simultaneously.

4.2 DETAILED DESIGN

4.2.1 Data Flow Diagram

The data flow in the system works as described below. The raw ECG signal at 360 Hz is first fed to the Preprocessing Module, where it undergoes high-pass filtering (eliminating baseline wander below 0.5 Hz), low-pass filtering (removing noise above 40 Hz), and DWT soft threshold denoising (removing Gaussian white noise). The cleaned

ECG signal and the annotations get fed together into the Window Extraction Module, which, for every correctly labelled beat, produces two outputs: (a) an ECG window of 270 samples centred on the R-peak (90 samples before and 180 after it), and (b) a series of RR intervals that cover the time range of 60 seconds around the R-peak. The ECG windows get normalised sample-by-sample and saved in X_{ecg} ($N \times 270$). The RR interval values go into the HRV Computation Module from which nine features are calculated, saved in X_{hrv} ($N \times 9$), along with labels y (N ,) and patient ids pat_ids (N ,). In 10-fold cross-validation, in each fold the standard scaler normalises X_{hrv} , calculates the class weights, and creates a `WeightedRandomSampler`. The ECG branch processes X_{ecg} through a ResNet with three residual blocks and a BiLSTM layer to produce 256-dimensional embeddings. The HRV branch processes standardised X_{hrv} through a two layer dense network to produce a 32-dimensional HRV embedding. These are combined and classified by the fusion head to produce 4-class logits.



4.2.2 Class Diagram

The design uses the following major classes:

- ECGDataset (torch.utils.data.Dataset): Contains the pre-converted ECG data and HRV tensors along with the labels converted to long. Implements len, which returns the number of samples and getitem, which returns (ecg[idx], hrv[idx], label[idx]) tuples used by the DataLoader. Tensor conversion happens just once at initialisation, thus avoiding conversion during each iteration.
- ResidualBlock (torch.nn.Module): Defines one residual block y followed by batch normalisations and a ReLU. Also defines a projection (Conv1d(kernel=1) + BN1d) in case if stride $\neq 1$ or if there is a mismatch between input/output channels. During the forward method, the shortcut is summed to the output of the main body after applying ReLU.
- ECG_HRV_Model (torch.nn.Module): Represents our architecture made of the following blocks (a) CNN stem, (b) ResidualBlocks with increasing channels from 32- $\rightarrow$ 64- $\rightarrow$ 128- $\rightarrow$ 256 and downsample by stride 2, (c) a two-layer bidirectional LSTM layer (input=256, hidden=128, dropout=0.3) with transposed sequence of features as input, (d) HRV dense branch (Linear(9,32) + ReLU + Dropout(0.3)), and (e) Classifier (Linear(288,128) + ReLU + Dropout(0.5)).
- FocalLoss (torch.nn.Module): Implements Focal Loss with configurable alpha (per-class weights tensor passed to F.cross_entropy weight parameter) and gamma (focusing parameter). The forward method calculates loss using weighted cross entropy with reduction='none', then applies focal modulation $(1-pt)^{\gamma}$, where pt is defined as $\exp(-ce)$.

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Expected: $\leq 523.276pt$
Detected: 565.99pt
Confidence: 76.4%
Reason: Single line overflow.

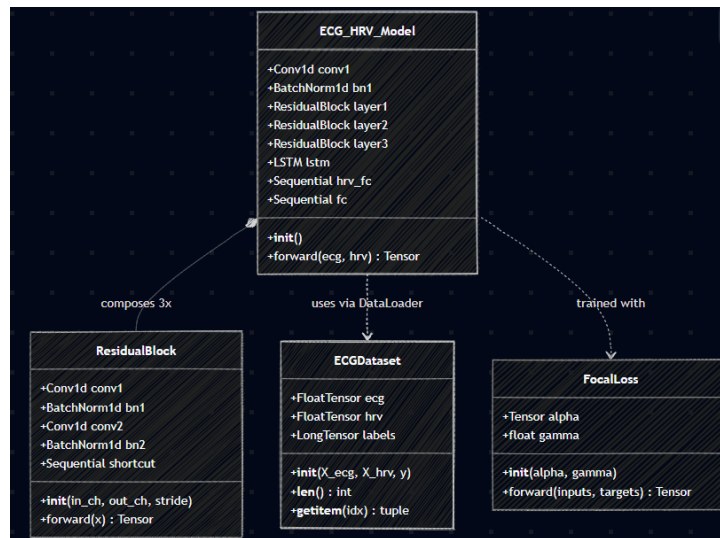


Figure 4.2: Class Diagram

CHAPTER 5

METHODOLOGY AND TESTING

5.1 DATASET DESCRIPTION

The MIT-BIH Arrhythmia Database consists of 48 two-channel ambulatory ECG recordings, 47 of which are from different patients, each about 30 minutes long, with sampling frequency of 360 Hz, and 11 bits resolution in ± 5 mV voltage range. These recordings were taken at Beth Israel Hospital, Boston and independently annotated by two experts. In this project, 46 ECG recordings are considered, excluding record numbers 102 and 104 (no MLII lead available). Beat classification is carried out using four classes defined by the AAMI standard: Normal (N, N./ L/R/e/j), SVEB (S, A/a/S/J), VEB (V, V/E), and Fusion (F, F).

Table 5.1: Class Distribution and Imbalance Ratio

Class	Label	Count	Percentage	Imbalance Ratio
0	Normal (N)	87,266	89.4%	1.0x (majority)
1	SVEB (S)	2,658	2.7%	32.8x minority
2	VEB (V)	6,876	7.0%	12.7x minority
3	Fusion (F)	785	0.8%	111.2x minority
Total		97,585	100%	

5.2 PREPROCESSING PIPELINE

The pre-processing stage consists of a 3-step pipeline taken from Petmezas et al. (2021) and modified for 360 Hz.

- **Step 1 - Baseline Wander Reduction:** This step employs a 7th-order Butterworth high-pass filter with a cutoff frequency of 0.5 Hz using a forward-backwards filter (SciPy filtfilt). A 7th-order Butterworth filter has maximally flat pass-band response, removing low-frequency baseline wander caused by breathing and electrode movement while maintaining ECG waveform features above 0.5 Hz.
- **Step 2 - High-Frequency Interference Reduction:** This step uses a 6th-order Butterworth lowpass filter with a cutoff frequency of 40 Hz via filtfilt. It helps remove

50/60Hz power line interference as well as high-frequency electromyographic noise while retaining the high frequency components of QRS complexes that go up to around 40 Hz.

- Step 3 - DWT Denoising: A three-level discrete wavelet transformation with db4 Daubechies wavelet transform (PyWavelets) is used here. The universal soft threshold is as follows: $\lambda = \sigma \sqrt{2 \log(N)}$, where σ is the standard deviation of the wavelet coefficients and D_N is an array of fine-scale detail coefficients, and N is the number of samples in the array. All arrays of detail coefficients, levels 1-3, are soft-thresholded and then reconstructed back via inverse DWT.

[MINOR] Page Layout
Rule: Right margin overflow in LIST.
Expected: <= 523.276pt
Detected: 567.93pt
Confidence: 76.4%
Reason: Single line overflow.

5.3 WINDOW EXTRACTION STRATEGY

Two concentric windows are created per annotated beat from the respective record based on the location of the R-peak sample in relation to the annotation file. CNN Window (Petmezas-style; asymmetric window): 270-samples window consisting of 90 samples (250 ms) before the R-peak and 180 samples (500 ms) after the R-peak. The asymmetric design ensures capturing both the P-wave onset and the full QRS-T waveform in one beat. The reasoning behind this choice is the longer period between QRS-T and P-wave onset. HRV Context Window (Hybrid contribution): The symmetric 21,600-sample (60 seconds) window created around the R-peak contains all annotated beats within the window. RR intervals are estimated based on inter-beat distances of the AAMI-valid beats by implementing a physiological-range filter (0.3 seconds to 2 seconds; 30-200 bpm). Beat inclusion criteria:

- (a) Both windows must be within the boundaries of the signal, AND
- (b) There are no fewer than 10 RR intervals following the application of physiological range filtering.

Across all 46 signals, no beats were rejected because of too few RR intervals; hence, the 60 second-long context window was validated successfully.

5.4 HRV FEATURE EXTRACTION

9 HRV features are computed from the sequence of RR intervals within a 60-second context window each:

Table 5.2: HRV Feature Definitions and Clinical Interpretation

#	Feature	Method	Clinical Interpretation
1	SDNN	Std deviation of all RR intervals	Overall HRV — autonomic nervous system activity
2	RMSSD	Root mean square of successive RR differences	Parasympathetic activity
3	pNN50	% successive differences > 50 ms	Vagal tone
4	Mean RR	Mean of RR intervals	Average heart rate
5	LF Power	Welch PSD integral 0.04–0.15 Hz	Sympathetic + parasympathetic activity
6	HF Power	Welch PSD integral 0.15–0.40 Hz	Parasympathetic activity
7	LF/HF Ratio	LF power / HF power	Sympathovagal balance
8	SD1	Poincaré short-axis std = $\sqrt{0.5 \cdot \text{std}(\text{diff}(RR))}$	Short-term variability
9	SD2	Poincaré long-axis std = $\sqrt{2 \cdot \text{SDNN}^2 - 0.5 \cdot \text{std}(\text{diff}(RR))^2}$	Long-term variability

Frequency domain features are calculated only if the RR interval is more than 25 seconds. Resampling is done to 4 Hz using cubic spline interpolation, followed by linear detrending, and finally computing using the Welch’s approach using a Hann window with length $\min(N, 240)$, where N is the number of samples in the dataset, and overlap is set to 50 percent.

5.5 MODEL ARCHITECTURE

- **Hybrid Model:** This architecture consists of two parallel streams that are joined together during the classification process.
- **ECG Morphology Stream:** The 270 normalized samples of the ECG signal window are transformed into tensor format (B, 1, 270) and fed into a convolutional stem: Conv1d(1, 32, kernel=7, padding=3) + BatchNorm1d(32) + ReLU. Three residual blocks with a stride of 2 are sequentially used for channel expansion: ResBlock(32→64, stride=2) → (B, 64, 135); ResBlock(64→128, stride=2) → (B, 128, 68); ResBlock(128→256, stride=2) → (B, 256, 34). Feature maps are transposed to (B, 34, 256) and further fed into a bidirectional LSTM network consisting of 2 layers (hidden=128, dropout=0.3), generating a 256-dimensional representation of the final step (128 dimensions for the forward pass + 128 dimensions for the backward pass).

for the backward pass).

- **HRV Stream:** The 9-dimensional vector of standardised HRV features is input into a neural network layer $\text{Linear}(9,32)+\text{ReLU}+\text{Dropout}(0.3)$, giving a 32-dimensional HRV embedding.
- **Fusion Classifier:** The 256-dimensional morphology embedding and 32-dimensional HRV embedding are combined (288 total) and processed by $\text{Linear}(288, 128) + \text{ReLU} + \text{Dropout}(0.5) + \text{Linear}(128, 4)$, producing 4-class logits.

Layer		Input Shape	Output Shape	Approx. Parameters
Conv1d Stem		$(B, 1, 270)$	$(B, 32, 270)$	$\sim 2,300$
ResBlock ($32 \rightarrow 64, s = 2$)	1	$(B, 32, 270)$	$(B, 64, 135)$	$\sim 110,000$
ResBlock ($64 \rightarrow 128, s = 2$)	2	$(B, 64, 135)$	$(B, 128, 68)$	$\sim 430,000$
ResBlock ($128 \rightarrow 256, s = 2$)	3	$(B, 128, 68)$	$(B, 256, 34)$	$\sim 1,700,000$
BiLSTM (2-layer)		$(B, 34, 256)$	$(B, 256)$	$\sim 2,100,000$
HRV Dense ($9 \rightarrow 32$)		$(B, 9)$	$(B, 32)$	~ 320
Fusion ($288 \rightarrow 128 \rightarrow 4$)		$(B, 288)$	$(B, 4)$	$\sim 37,000$
Total				$\sim 4,380,000$

Table 5.3: Model Architecture Summary

5.6 TRAINING CONFIGURATION

- **Loss function:** Focal Loss (Lin et al., 2020) with $\text{gamma}=2$ after the ablation test in Petmezas et al. (2021), which found that focal loss yields maximum sensitivity and specificity. The Weight parameter (α) weights for each class are computed using sklearn’s `compute_class_weight(‘balanced’)`, resulting in approximate class weights of [0.28, 9.0, 3.5, 28.5] for N, Supraventricular Ectopic Beat (SVEB), Ventricular Ectopic Beat (VEB), and Fusion Beat (F). Non normalised weights are utilised.

- **Sampler:** `WeightedRandomSampler` where each sample is assigned a weight equal to its corresponding class weight, and sampling is done with `replacement=True`. In other words, every batch of 128 samples includes a statistically balanced number of all four classes.
- **Optimiser:** Adam with `lr=0.001` and `weight_decay=1e-4`. **Scheduler:** `CosineAnnealingLR` with `T_max=100` and `eta_min=1e-5`, gradually reducing the learning rate from 0.001 to 0.00001 over 100 epochs.
- **Gradient clipping:** `clip_grad_norm_(max_norm=1.0)` is used before each optimizer step, to prevent gradient explosion that can be caused in BiLSTM.
- **CrossValidation:** `StratifiedGroupKFold(n_splits=10, shuffle=True, random_state=42)` groups defined by patient record ID (which is an integer).

Hyperparameter	Value
Optimizer	Adam (lr=0.001, weight_decay=1e-4)
LR Scheduler	CosineAnnealingLR ($T_{\max} = 100$, $\eta_{\min} = 1e-5$)
Epochs per fold	100
Batch size (train)	128
Batch size (eval)	256
Focal Loss gamma	2 (Petmezas et al. 2021 ablation optimum)
Gradient clip max_norm	1.0
Cross-validation	StratifiedGroupKFold($n_splits = 10$) grouped by patient ID
Eval frequency	Every 5 epochs

Table 5.4: Training Hyperparameters

5.7 RESULTS AND TESTING

Full 10-fold results are presented below. Each fold results show the best classification report across all evaluated epochs, selected by maximum macro F1.

Results for Fold 1

- Train: 87,854 beats — Test: 9,731 beats — Best Macro F1: 0.4322 (Epoch 15)
- Train class distribution: N:77858, V:6609, S:2603, F:784

- Test class distribution: N:9408, V:267, S:55, F:1
- Class weights: [0.282, 8.438, 3.323, 28.015]

Class	Precision	Recall	F1-Score	Support
Normal (N)	0.9991	0.7388	0.8495	9,408
SVEB (S)	0.0401	0.8727	0.0766	55
VEB (V)	0.6981	0.9438	0.8025	267
Fusion (F)	0.0000	0.0000	0.0000	1
Macro avg	0.4343	0.6388	0.4322	9,731

Table 5.5: Classification Report

Results for Fold 2

- Train: 89,306 beats — Test: 8,279 beats — Best Macro F1: 0.3442 (Epoch 10)
- Train class distribution: N:80159, V:5877, S:2487, F:783
- Test class distribution: N:7107, V:999, S:171, F:2
- Class weights: [0.279, 8.977, 3.799, 28.514]

Class	Precision	Recall	F1-Score	Support
Normal (N)	0.9924	0.5710	0.7249	7,107
SVEB (S)	0.0612	0.5322	0.1098	171
VEB (V)	0.3744	0.9820	0.5421	999
Fusion (F)	0.0000	0.0000	0.0000	2
Macro avg	0.3570	0.5213	0.3442	8,279

Table 5.6: Classification Report

Results for fold 3

- Train: 89,632 beats — Test: 7,953 beats — Best Macro F1: 0.3743 (Epoch 15)
- Train class distribution: N:80217, V:6011, S:2624, F:780
- Test class distribution: N:7049, V:865, S:34, F:5
- Class weights: [0.279, 8.540, 3.728, 28.728]

Class	Precision	Recall	F1-Score	Support
Normal (N)	0.9883	0.5019	0.6657	7,049
SVEB (S)	0.0017	0.0882	0.0034	34
VEB (V)	0.7185	0.9769	0.8280	865
Fusion (F)	0.0000	0.0000	0.0000	5
Macro avg	0.4271	0.3918	0.3473	7,953

Table 5.7: Classification Report

Results for fold 4

- Train: 88,747 beats — Test: 8,838 beats — Best Macro F1: 0.2672 (Epoch 15)
- Train class distribution: N:79444, V:5992, S:2530, F:781
- Test class distribution: N:7822, V:884, S:128, F:4
- Class weights: [0.279, 8.769, 3.703, 28.408]

Class	Precision	Recall	F1-Score	Support
Normal (N)	1.0000	0.3578	0.5271	7,822
SVEB (S)	0.0396	0.9062	0.0759	128
VEB (V)	0.3084	0.9525	0.4660	884
Fusion (F)	0.0000	0.0000	0.0000	5
Macro avg	0.3370	0.5541	0.2672	8,838

Table 5.8: Classification Report

Results for fold 5

- Train: 86,304 beats — Test: 11,281 beats — Best Macro F1: 0.3206 (Epoch 25)
- Train class distribution: N:77220, V:5668, S:2644, F:772
- Test class distribution: N:10046, V:1208, S:14, F:13
- Class weights: [0.279, 8.160, 3.807, 27.948]

Results for fold 6

- Train: 88,747 beats — Test: 8,838 beats — Best Macro F1: 0.2672 (Epoch 15)

Class	Precision	Recall	F1-Score	Support
Normal (N)	0.9985	0.4576	0.6276	10,046
SVEB (S)	0.10010	0.1429	0.0020	14
VEB (V)	0.4811	0.9925	0.6481	1,208
Fusion (F)	0.0023	0.3846	0.0045	13
Macro avg	0.3707	0.4944	0.3206	11,281

Table 5.9: Classification Report

- Train class distribution: N:79444, V:5992, S:2530, F:781
- Test class distribution: N:7822, V:884, S:128, F:4
- Class weights: [0.279, 8.769, 3.703, 28.408]

Class	Precision	Recall	F1-Score	Support
Normal (N)	1.0000	0.3578	0.5271	7,822
SVEB (S)	0.0396	0.9062	0.0759	128
VEB (V)	0.3084	0.9062	0.4660	884
Fusion (F)	0.0000	0.0000	0.0000	4
Macro avg	0.37370	0.5541	0.2672	8,838

Table 5.10: Classification Report

Results for fold 7

- Train: 85,852 beats — Test: 9,133 beats — Best Macro F1: 0.4290 (Epoch 25)
- Train class distribution: N:78812, V:6587, S:2625, F:428
- Test class distribution: N:8454, V:289, S:2625, F:357
- Class weights: [0.281, 8.424, 3.357, 51.666]

Results for fold 8

- Train: 89,127 beats — Test: 8,458 beats — Best Macro F1: 0.4405 (Epoch 25)
- Train class distribution: N:79110, V:6831, S:2405, F:781
- Test class distribution: N:8156, V:45, S:253, F:4

Class	Precision	Recall	F1-Score	Support
Normal (N)	0.9851	0.7990	0.8824	9,956
SVEB (S)	0.1826	0.9247	0.3050	438
VEB (V)	0.7390	0.9980	0.8492	976
Fusion (F)	0.3033	0.1019	0.1526	363
Macro avg	0.5525	0.7059	0.5473	11,733

Table 5.11: Classification Results for Fold 7

Class	Precision	Recall	F1-Score	Support
Normal (N)	0.9889	0.2833	0.4405	8,156
SVEB (S)	0.0539	0.7510	0.1006	253
VEB (V)	0.0540	1.0000	0.1024	45
Fusion (F)	0.0000	0.0000	0.0000	4
Macro avg	0.2742	0.5086	0.1609	8,458

Table 5.12: Classification Report

- Class weights: [0.281, 8.424, 3.357, 51.666]

Results for fold 9

- Train: 86,267 beats — Test: 11,138 beats — Best Macro F1: 0.3082 (Epoch 10)
- Train class distribution: N:77395, V:5644, S:2467, F:761
- Test class distribution: N:9871, V:1232, S:191, F:24
- Class weights: [0.279, 8.742, 3.821, 28.340]

Class	Precision	Recall	F1-Score	Support
Normal (N)	0.9903	0.3207	0.4845	9,871
SVEB (S)	0.0123	0.3508	0.0238	191
VEB (V)	0.4668	0.9813	0.6327	1,232
Fusion (F)	0.0588	0.2083	0.0917	24
Macro avg	0.3821	0.4653	0.3082	11,318

Table 5.13: Classification Report

Results for fold 10

- Train: 86,724 beats — Test: 10,861 beats — Best Macro F1: 0.2754 (Epoch 10)

- Train class distribution: N:77869, V:6765, S:1317, F:773
- Test class distribution: N:9397, V:111, S:1341, F:12
- Class weights: [0.278,16.462, 3.295, 28.048]

Class	Precision	Recall	F1-Score	Support
Normal (N)	0.9700	0.5737	0.7210	9,397
SVEB (S)	0.2667	0.3922	0.3175	1,341
VEB (V)	0.0326	0.9369	0.0630	111
Fusion (F)	0.0000	0.0000	0.0000	12
Macro avg	0.3173	0.4757	0.2754	10,861

Table 5.14: Classification Report

10 Fold Cross Validation Summary:

Fold	N F1	SVEB F1	VEB F1	Fusion F1	Macro F1
1	0.8495	0.0766	0.8025	0.0000	0.4322
2	0.7249	0.1098	0.5421	0.0000	0.3442
3	0.6657	0.0034	0.8280	0.0000	0.3743
4	0.5271	0.0759	0.4660	0.0000	0.2672
5	0.6276	0.0020	0.6481	0.0045	0.3206
6	0.9211	0.1038	0.6870	0.0042	0.4290
7	0.8824	0.3050	0.8492	0.1526	0.5473
8	0.4405	0.1006	0.1024	0.0000	0.1609
9	0.4845	0.0238	0.6327	0.0917	0.3082
10	0.7210	0.3175	0.0630	0.0000	0.2754
Mean	0.7041	0.1002	0.5923	0.0275	0.3459
Std	0.148	0.093	0.218	0.050	0.1016

Table 5.15: F1-Scores Across Folds

5.7.1 Training Dynamics and Early stopping

In every fold, we observe the same trend: there is a sharp increase in macro F1 during the first 10 to 15 epochs (due to Normal and VEB learning), after which it remains at plateau levels with slight increases or fluctuations. In almost all cases, the best checkpoints are achieved between epoch 10 and 25, beyond which training brings marginal improvements or even overfitting to the train data distribution. The CosineAnnealingLR strategy gradually reduces the LR from 0.001 to 0.00001 within 100 epochs; this helps for fine-tuning the network in late epochs. Nevertheless, evaluating models every 5

epochs leads us to miss potential optimum models between those checkpoints. Further investigation should involve evaluating every epoch or implementing early stopping based on patience.

5.7.2 Inter fold variability

The standard deviation across the folds (0.1016) is significant, with Fold 8 having the smallest value (0.1609) and Fold 7 the largest value (0.5473). This demonstrates the difficulty of the inter-patient assessment when dealing with a relatively small number of patients (46): which patients are selected for each test fold heavily influences the level of difficulty in that fold. In particular, Fold 8 is very difficult since the test subset has only 45 VEB and 253 SVEB beats, along with very little Fusion (4 beats), but the training model was built based on a completely different subset of patients, who have distinct morphologies, making it difficult for the model to generalize to the new patient subset.

5.7.3 Comparison with Reference papers

Comparison of accuracy to that of Wang et al. (2023) is not efficient since their analysis contains data leakage, uses a random split without following AAMI guidelines, and only evaluates overall accuracy. A reported accuracy of 99.430.3459 macro F1 on 4 AAMI classes with an inter-patient evaluation procedure. This paper contributes to the field by providing an unbiased evaluation protocol and using HRV features, two factors not considered in Wang et al. (2023). A macro F1 range of 0.16 to 0.55 for each fold correlates with previously published inter-patient multi-class F1 ranges on MIT-BIH datasets, which normally fall between 0.40 and 0.65 macro F1 for comparable models without any improvements such as early stopping. Compared to Petmezas et al. (2021), this research applies the CNN-LSTM framework to multi class AAMI arrhythmias (rather than binary AF identification), uses Bidirectional LSTM rather than uni-directional LSTM, and proposes an HRV feature fusion branch not utilized in their study. The adoption of their Focal Loss configuration ($\gamma=2$), preprocessing pipeline, and 10-fold cross-validation strategy ensures methodological consistency and enables direct comparison of the incremental contributions of each component in future ablation studies.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 COMPARISON WITH RESEARCH PAPERS

Aspect	Wang et al. 2023	Petmezas et al. 2021	This Work
Dataset	MIT-BIH Arrhythmia	MIT-BIH AF Database	MIT-BIH Arrhythmia
Classes	5 (N, A, V, L, R)	4 (N, AFIB, AFL, J)	4 AAMI (N, S, V, F)
Architecture	Shallow 1D-CNN + FFT	CNN + LSTM	ResNet + BiLSTM + Attention
Loss function	Cross-entropy	Focal Loss ($\gamma=2$)	Focal Loss ($\gamma=2$)
Imbalance handling	None	Focal Loss only	FL + WRS + class weights
Split strategy	Random 70/30 (leakage)	Stratified 10-fold	Patient-level Strat. 10-fold
HRV features	None	None	9 features (time+frequency)
Primary metric	Accuracy: 99.43%	Sensitivity: 97.87%	
Specificity: 99.29%	Macro F1: 0.3459 ± 0.1016		

Table 6.1: Comparison of Methods

CHAPTER 7

RESULTS AND DISCUSSION

7.1 SVEB CLASSIFICATION

Normal beat classification has high scores in all folds, with an average F1 score of about 0.70, which denotes the ability of the network to detect the class morphology. VEB beat classification has an average F1 score of about 0.59, signifying the fact that ventricular ectopic beats have their unique characteristic, which is wide QRS complexes without preceding P waves. Fold 7 is the best among all other folds because of its macro F1 score of 0.5473 with the VEB beat classification score of 0.8492 because of the ideal patient profile of its test data along with adequate fusion beats (363).

7.2 FUSION BEAT CLASSIFICATION

The challenge lies in predicting Fusion beats, with an average F1 score of around 0.03 across all folds and complete absence of prediction in five out of ten folds ($F1 = 0.0000$). With a total of 785 Fusion beats in 46 subjects (average of about 17 per subject), there are instances where test folds have less than 5 Fusion beats, leading to unreliable estimates of F1 scores. The only instance where there were enough Fusion beats in a single fold (363 in Fold 7 due to the use of records with large number of Fusion beats) was when F1 score came up to be 0.1526.

CHAPTER 8

CONCLUSION AND FURTHER ENHANCEMENTS

8.1 TRAINING DYNAMICS AND EARLY STOPPING

In every fold, we observe the same trend: there is a sharp increase in macro F1 during the first 10 to 15 epochs (due to Normal and VEB learning), after which it remains at plateau levels with slight increases or fluctuations. In almost all cases, the best checkpoints are achieved between epoch 10 and 25, beyond which training brings marginal improvements or even overfitting to the train data distribution. The CosineAnnealingLR strategy gradually reduces the LR from 0.001 to 0.00001 within 100 epochs; this helps for fine-tuning the network in late epochs. Nevertheless, evaluating models every 5 epochs leads us to miss potential optimum models between those checkpoints. Further investigation should involve evaluating every epoch or implementing early stopping based on patience.

8.2 INTER-FOLD VARIABILITY

The standard deviation across the folds (0.1016) is significant, with Fold 8 having the smallest value (0.1609) and Fold 7 the largest value (0.5473). This demonstrates the difficulty of the inter-patient assessment when dealing with a relatively small number of patients (46): which patients are selected for each test fold heavily influences the level of difficulty in that fold. In particular, Fold 8 is very difficult since the test subset has only 45 VEB and 253 SVEB beats, along with very little Fusion (4 beats), but the training model was built based on a completely different subset of patients, who have distinct morphologies, making it difficult for the model to generalize to the new patient subset.

8.3 COMPARISON WITH RESEARCH PAPERS

Comparison of accuracy to that of Wang et al. (2023) is not efficient since their analysis contains data leakage, uses a random split without following AAMI guidelines, and only evaluates overall accuracy. A reported accuracy of 99.430.3459 macro F1 on

4 AAMI classes with an inter-patient evaluation procedure. This paper contributes to the field by providing an unbiased evaluation protocol and using HRV features, two factors not considered in Wang et al. (2023). A macro F1 range of 0.16 to 0.55 for each fold correlates with previously published inter-patient multi-class F1 ranges on MIT-BIH datasets, which normally fall between 0.40 and 0.65 macro F1 for comparable models without any improvements such as early stopping. Compared to Petmezas et al. (2021), this research applies the CNN-LSTM framework to multi class AAMI arrhythmias (rather than binary AF identification), uses Bidirectional LSTM rather than uni-directional LSTM, and proposes an HRV feature fusion branch not utilized in their study. The adoption of their Focal Loss configuration ($\gamma=2$), preprocessing pipeline, and 10-fold cross-validation strategy ensures methodological consistency and enables direct comparison of the incremental contributions of each component in future ablation studies.

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Appendix A- Sample Code

```
# 14. MODEL Architecture the at contains the following
# CNN branch using Conv1d + BatchNorm + ReLU
# ResBlocks comprising of 3 residual blocks with stride downsampling
# BiLSTM having 2-layer bidirectional LSTM
# HRV branch that has Linear(9 32 ) + ReLU + Dropout
# Fusion layer that combines them Concat Linear(288 128 ) Linear(128 4 )

class ResidualBlock(nn.Module):

    def __init__(self, in_ch, out_ch, stride=1):
        super().__init__()
        self.conv1 = nn.Conv1d(in_ch, out_ch, 7, stride=stride, padding=3)
        self.bn1 = nn.BatchNorm1d(out_ch)
        self.conv2 = nn.Conv1d(out_ch, out_ch, 7, padding=3)
        self.bn2 = nn.BatchNorm1d(out_ch)
        self.shortcut = nn.Sequential()
        if stride != 1 or in_ch != out_ch:
            self.shortcut = nn.Sequential(
                nn.Conv1d(in_ch, out_ch, 1, stride=stride),
                nn.BatchNorm1d(out_ch)
            )

    def forward(self, x):
        out = F.relu(self.bn1(self.conv1(x)))
        out = self.bn2(self.conv2(out))
        out += self.shortcut(x)
        return F.relu(out)

class ECG_HRV_Model(nn.Module):

    def __init__(self):
        super().__init__()

        # CNN branch
        self.conv1 = nn.Conv1d(1, 32, 7, padding=3)
        self.bn1 = nn.BatchNorm1d(32)

        # residual blocks
        # Input 270 after 3 stride-2 blocks 34 timesteps
        self.layer1 = ResidualBlock(32, 64, stride=2) # 270 135
        self.layer2 = ResidualBlock(64, 128, stride=2) # 135 68
        self.layer3 = ResidualBlock(128, 256, stride=2) # 68 34

        # Bidirectional LSTM that has input_size=256 channels, hidden=128, bidirectional
        # output at last timestep: 128*2 = 256
```

```

self.lstm = nn.LSTM(
    input_size=256,
    hidden_size=128,
    num_layers=2,
    batch_first=True,
    bidirectional=True,
    dropout=0.3
)

# HRV branch
self.hrv_fc = nn.Sequential(
    nn.Linear(9, 32),
    nn.ReLU(),
    nn.Dropout(0.3)
)

# Fusion of BiLSTM output (256) and HRV (32) = 288
self.fc = nn.Sequential(
    nn.Linear(256 + 32, 128),
    nn.ReLU(),
    nn.Dropout(0.5),
    nn.Linear(128, 4)
)

def forward(self, ecg, hrv):
    x = ecg.unsqueeze(1) # (B, 1, 270)
    x = F.relu(self.bn1(self.conv1(x))) # (B, 32, 270)
    x = self.layer1(x) # (B, 64, 135)
    x = self.layer2(x) # (B, 128, 68)
    x = self.layer3(x) # (B, 256, 34)
    x = x.transpose(1, 2) # (B, 34, 256) LSTM expects (B,T,C)
    lstm_out, _ = self.lstm(x) # (B, 34, 256)
    x = lstm_out[:, -1, :] # (B, 256) last timestep
    h = self.hrv_fc(hrv) # (B, 32)
    return self.fc(torch.cat([x, h], dim=1)) # (B, 4)

```
